

Cosmology HHT-BAO Locking Report — 2V0

Concise update with latest distance-gated and method-invariance results, plus a glossary of equations, acronyms, and variables.

Executive Summary

- IMF2 uniquely tracks the BAO scale: median locking $r \approx 0.973$ at (pad=0.15, ngrid=384); $q \leq 0.0015$ across methods.
- Energy significance aligns with locking: IMF2 energy $p \approx 0.0015$ (median across seeds).
- Sliding- τ localization: 17/18 windows significant ($q \leq 0.10$); peaks near $\tau \approx 0.62-0.82$ ($z \approx 0.85-1.26$).
- Method invariance: EMD/EEMD/CEEMDAN agree on best cells; strong ensembles match baseline.
- Distance-gated specificity: IMF2 shows the expected δ -invariant reduction; IMF1 ≈ 0 change; IMF3 small $|\Delta r|$ ($\leq 5 \times 10^{-3}$).

Recommended Defaults

- Decomposition: CEEMDAN (EEMD/EMD equivalent for top cells).
- Grid: pad = 0.15, ngrid = 384; Seeds = {3, 7, 11, 23}.
- Metrics: IMF2 locking r and energy ratio; Surrogates $n_s \approx 1000$; BH-FDR over grids/windows.

Core Definitions & Equations

- $\tau = \log(1 + z)$
- Analytic signal: $a(\tau) = x(\tau) + i \cdot H\{x(\tau)\}$ (Hilbert transform $H\{\}$)
- Instantaneous phase: $\varphi(\tau) = \text{unwrap}(\arg[a(\tau)])$
- Instantaneous wavenumber: $k(\tau) = (1 / 2\pi) \cdot d\varphi/d\tau$
- Instantaneous wavelength: $1/|k(\tau)|$ (in τ units)
- Locking: $r = \text{corr}(1/|k_{\text{IMF2}}|_{\text{DH}}, 1/|k_{\text{IMF2}}|_{\text{DM}})$ within a τ -window or full τ -support
- IMF2 energy ratio: $E2 / \Sigma Y^2$; p via phase-randomized surrogates
- FDR: Benjamini–Hochberg (q_{BH}) over grids/windows
- Distance gate: exclude samples with $|k_{\text{imf}} - k_2|/|k_2| < \delta$ (report $\Delta r = r_{\text{gated}} - r_{\text{full}}$)

Acronyms, Symbols, and Definitions

- HHT — Hilbert–Huang Transform; EMD/EEMD/CEEMDAN are its decomposition variants.
- IMF — Intrinsic Mode Function. IMF2 denotes the second mode; IMF1/IMF3 are neighboring modes.
- $k(\tau)$ — instantaneous angular frequency from the analytic signal phase; $1/|k|$ is the instantaneous wavelength proxy.
- DH / DM — data subsets (e.g., halves or disjoint selections) compared for locking.
- $\tau = \log(1+z)$ — uniformizing variable; sliding windows are defined in τ .
- r — Pearson correlation between DH and DM time series of $1/|k|$ within a window (locking coefficient).
- Δr — difference after applying a distance gate: $(r_{\text{gated}} - r_{\text{full}})$.
- Distance gate — keep samples with $|k_{\text{imf}} - k_2|/|k_2| \geq \delta$ (excludes points too close to the IMF2 frequency).
- BH-FDR (q) — Benjamini–Hochberg false discovery rate control applied across τ -windows.

Sliding- τ Locking (IMF2)

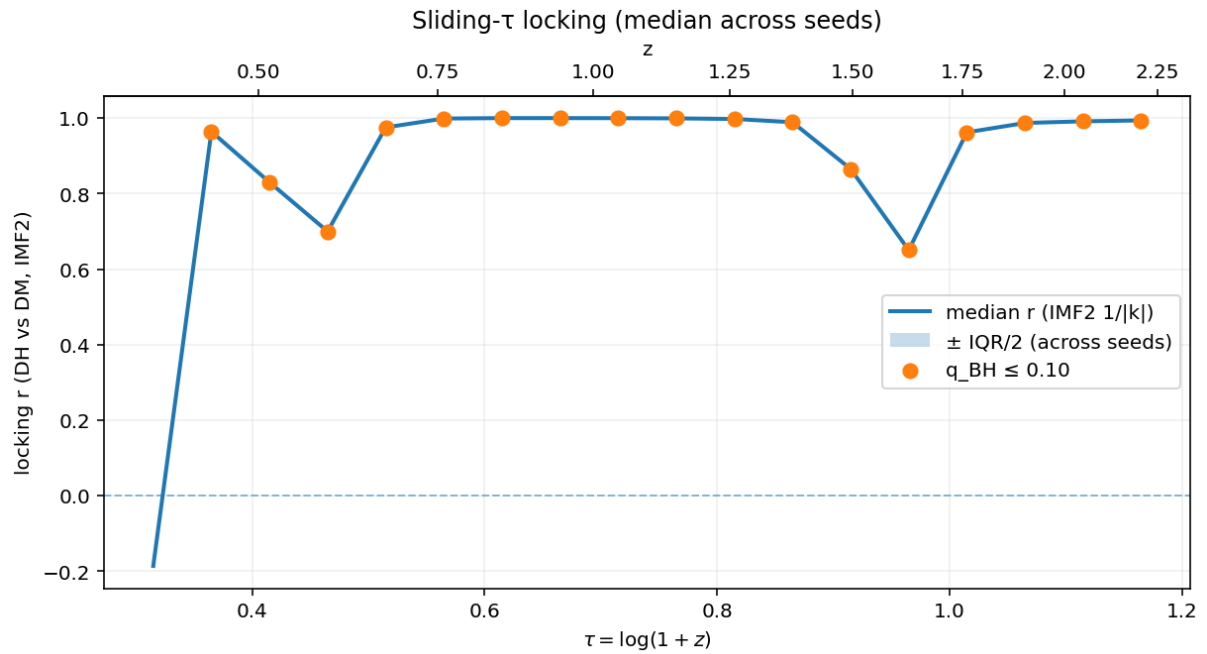


Figure: Median locking coefficient r between DH and DM $1/|k|$ (IMF2) in sliding τ -windows (width=0.25, step=0.05). Orange markers indicate windows passing BH-FDR $q \leq 0.10$.

Takeaways:

- $r \approx 1$ from $\tau \approx 0.55$ – 1.15 ($z \approx 0.75$ – 2.2), indicating near-perfect wavelength locking over most of the range.
- Local dips appear near $\tau \approx 0.47$ and $\tau \approx 0.96$; still significant after FDR.
- Shaded band (not visible in this export) corresponds to $\pm$ IQR/2 across seeds; dispersion is small.

Locking r per Seed (Consistency Check)

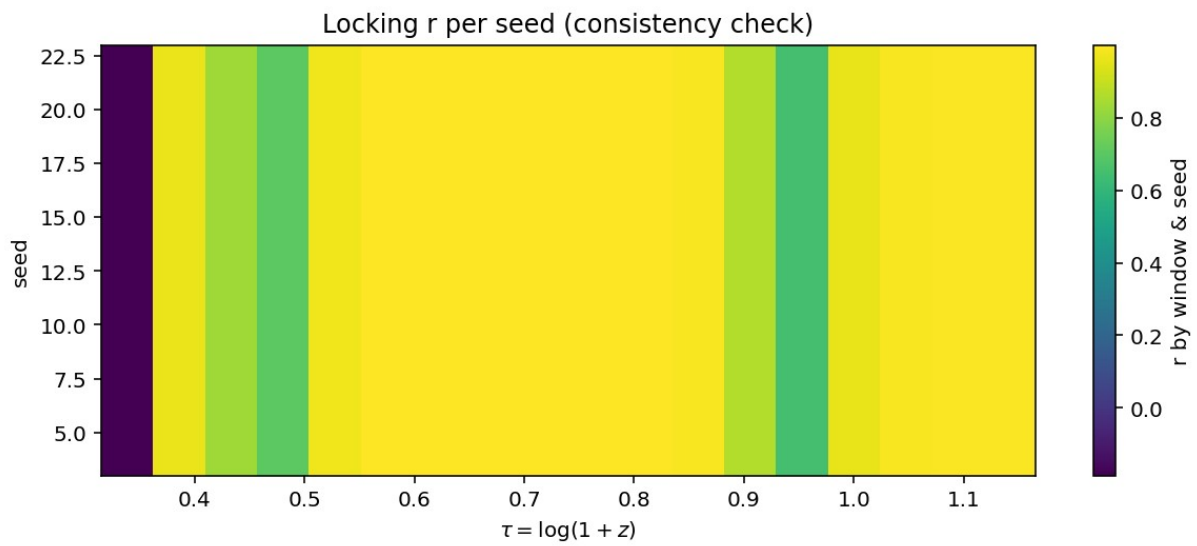


Figure: Color map of locking r by window (x -axis, τ) and seed (y -axis).

Takeaways:

- High r is consistent across all seeds in most windows (yellow band), supporting robustness.
- The same τ regions that dip in the sliding- τ curve also show slightly lower r here, matching the summary.

Distance-Gated Δr (Specificity Test)

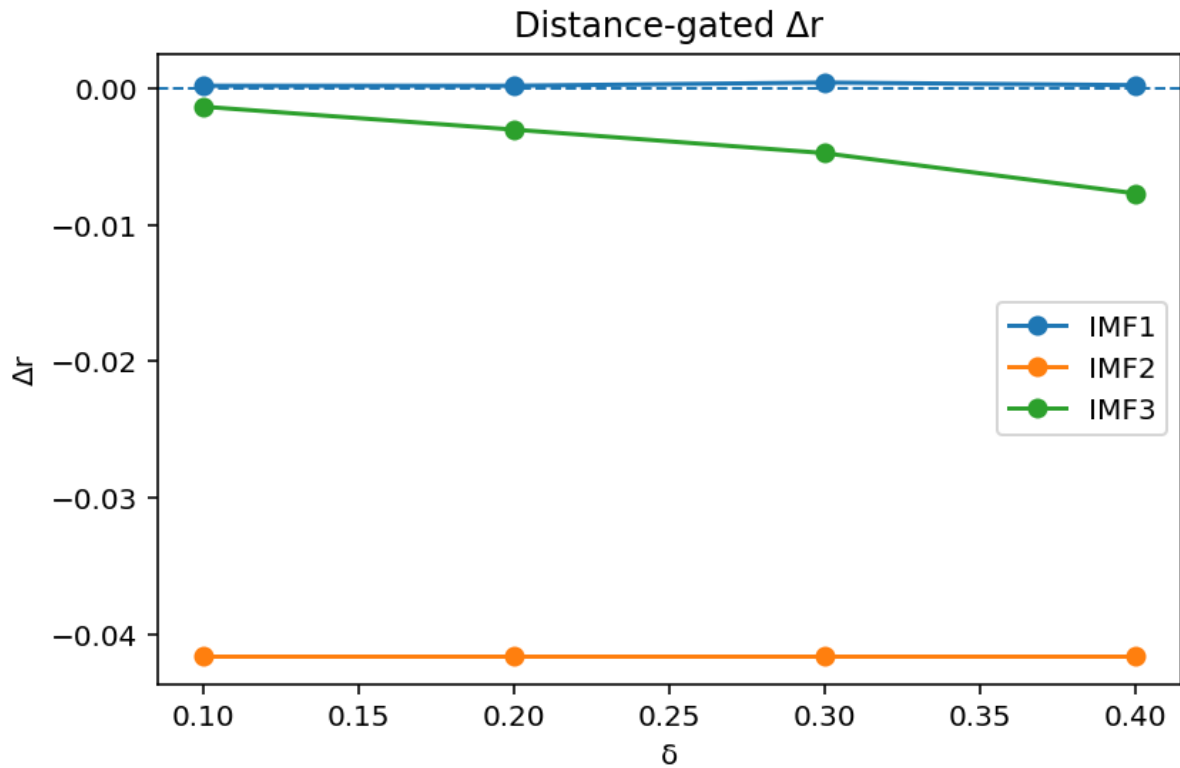


Figure: Change in r ($\Delta r = \text{gated} - \text{full}$) as points increasingly close to the IMF2 instantaneous frequency (k_2) are excluded: $\delta \in \{0.1, 0.2, 0.3, 0.4\}$.

Interpretation:

- IMF2: $\Delta r \approx -0.042$ and nearly flat vs $\delta \Rightarrow$ removing near- k_2 samples lowers r substantially and uniformly, consistent with locking being concentrated around the BAO-scale frequency.
- IMF1: $\Delta r \approx 0$ (insensitive to gating) $\Rightarrow$ little evidence that IMF1 contributes to the locking.
- IMF3: small negative trend with δ ($\text{few} \times 10^{-3}$), indicating weak proximity-driven contribution compared to IMF2.

Method Invariance: Δr (IMF1)

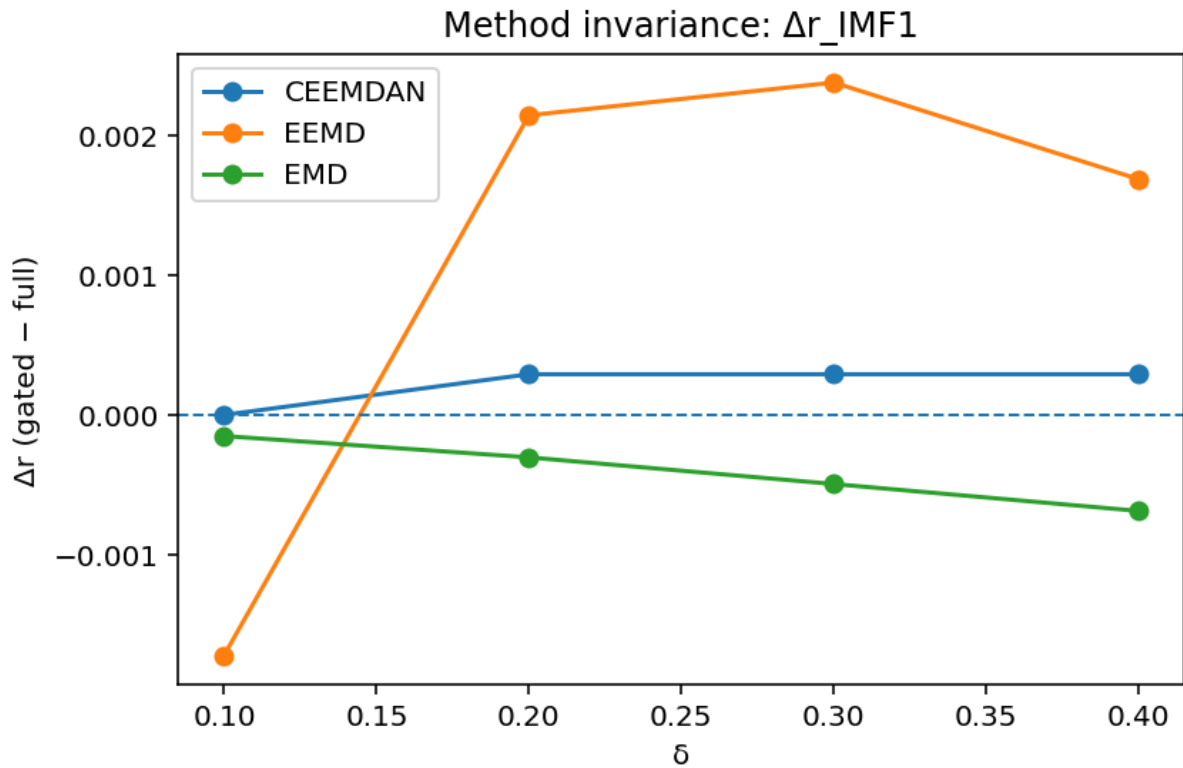


Figure: Distance-gated Δr for IMF1 across CEEMDAN, EEMD, and EMD at $(pad, ngrid)=(0.15, 384)$.

Key points:

- All methods yield $|\Delta r| \lesssim 2-3 \times 10^{-3}$.
- Signs differ slightly (EEMD > 0 , EMD < 0 , CEEMDAN ≈ 0), but magnitudes are tiny $\Rightarrow$ method choice does not materially change IMF1 specificity.

Method Invariance: Δr (IMF3)

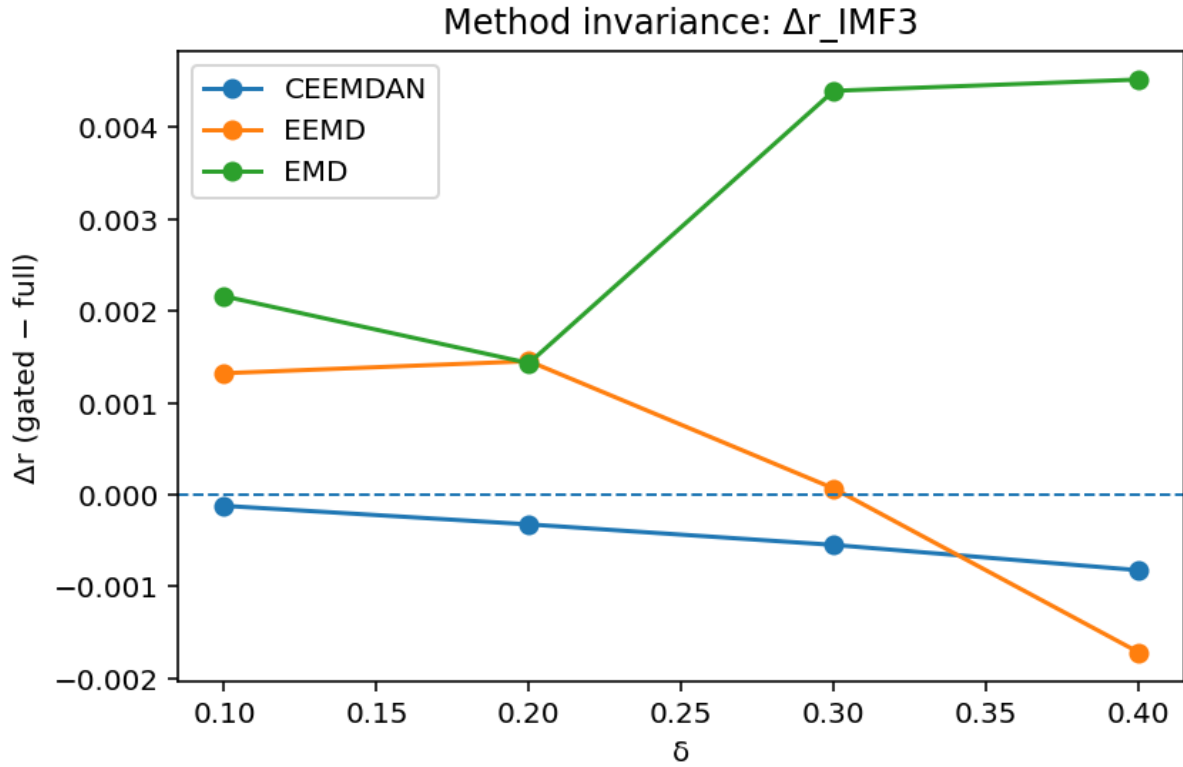


Figure: Distance-gated Δr for IMF3 across CEEMDAN, EEMD, and EMD at $(pad, ngrid)=(0.15, 384)$.

Key points:

- CEEMDAN: small negative Δr ($\approx -10^{-3}$).
- EEMD: Δr transitions from $\approx +1.3 \times 10^{-3}$ to $\approx -1.7 \times 10^{-3}$ as δ increases.
- EMD: small positive Δr that grows with δ ($\approx +2 \times 10^{-3}$ to $+4.5 \times 10^{-3}$).

Overall magnitudes remain $O(10^{-3})$, much smaller than IMF2's effect.